

Late Delivery Prediction for E-Commerce Orders (Brazil)

Group 10

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Business Problem



High volume: 96,476 orders processed in 2 years



SLA needs improvement: 27% orders were late deliveries (i.e. did not meet 14-day promise)



Monetary impact: R\$532,612 paid in compensation



Long term impact: Negative customer experience

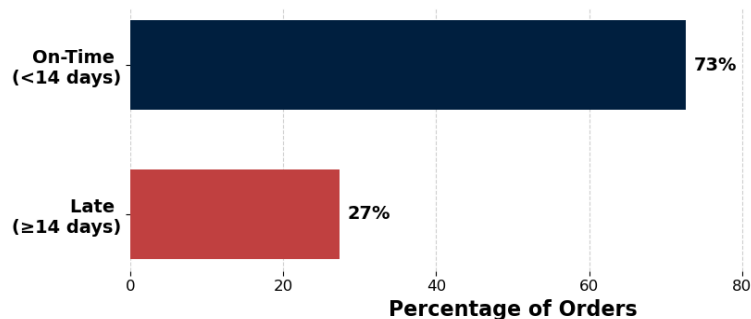
Business Goals

- Reduce from 27% to 20% for late orders
- Would save us R\$138,085 in paid compensation

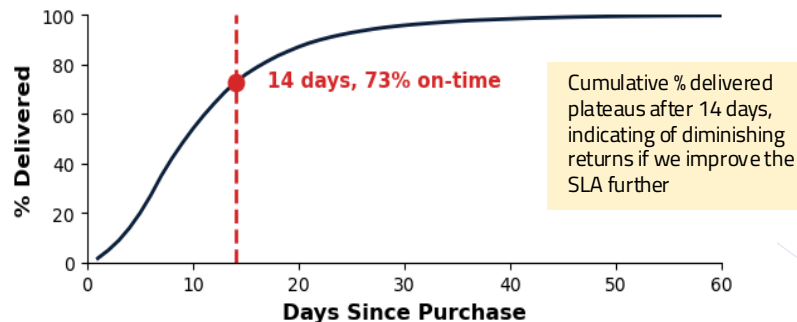
Proposed Solution

- Machine Learning model that predicts delivery status
- Target: 65% F1-score (min 26% to hit biz goal)

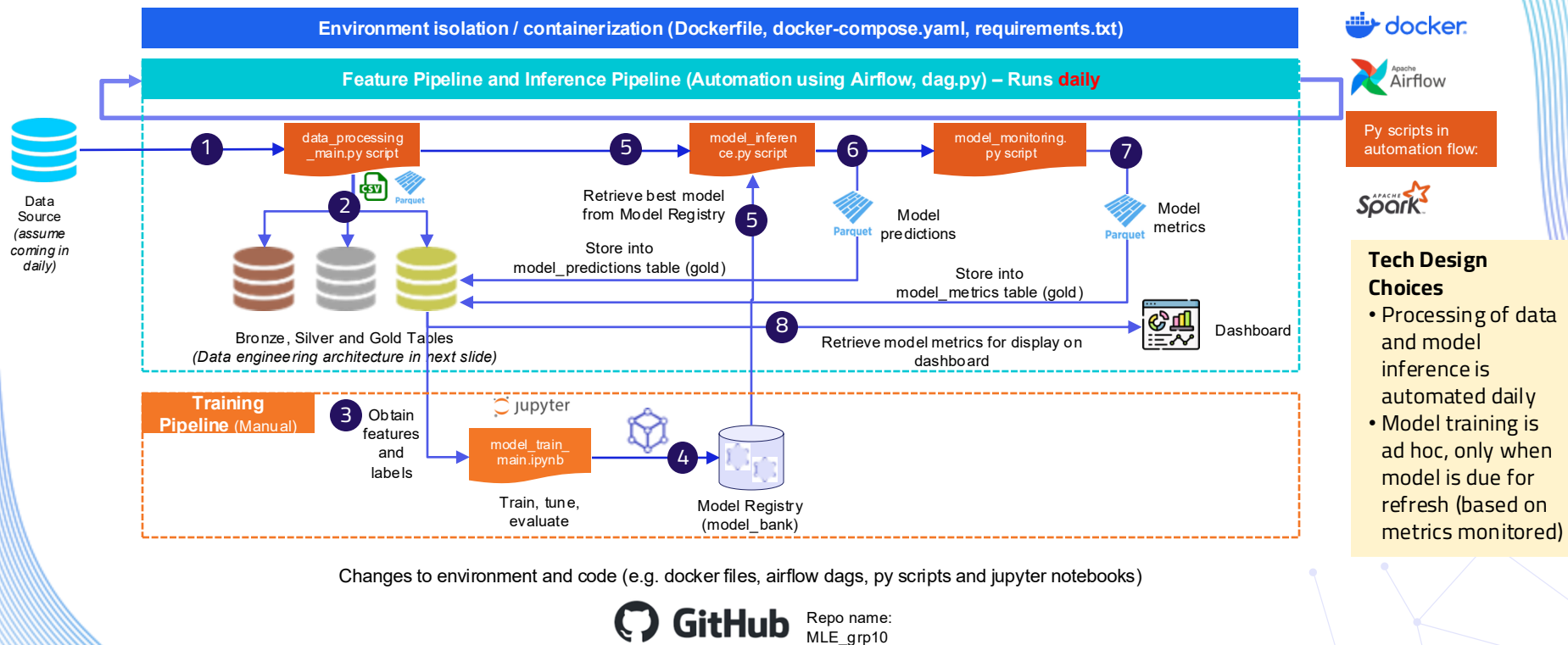
Percentage of On-Time vs. Late Deliveries
September 2016 - August 2018



Percentage of Orders Delivered On Time Since Purchase

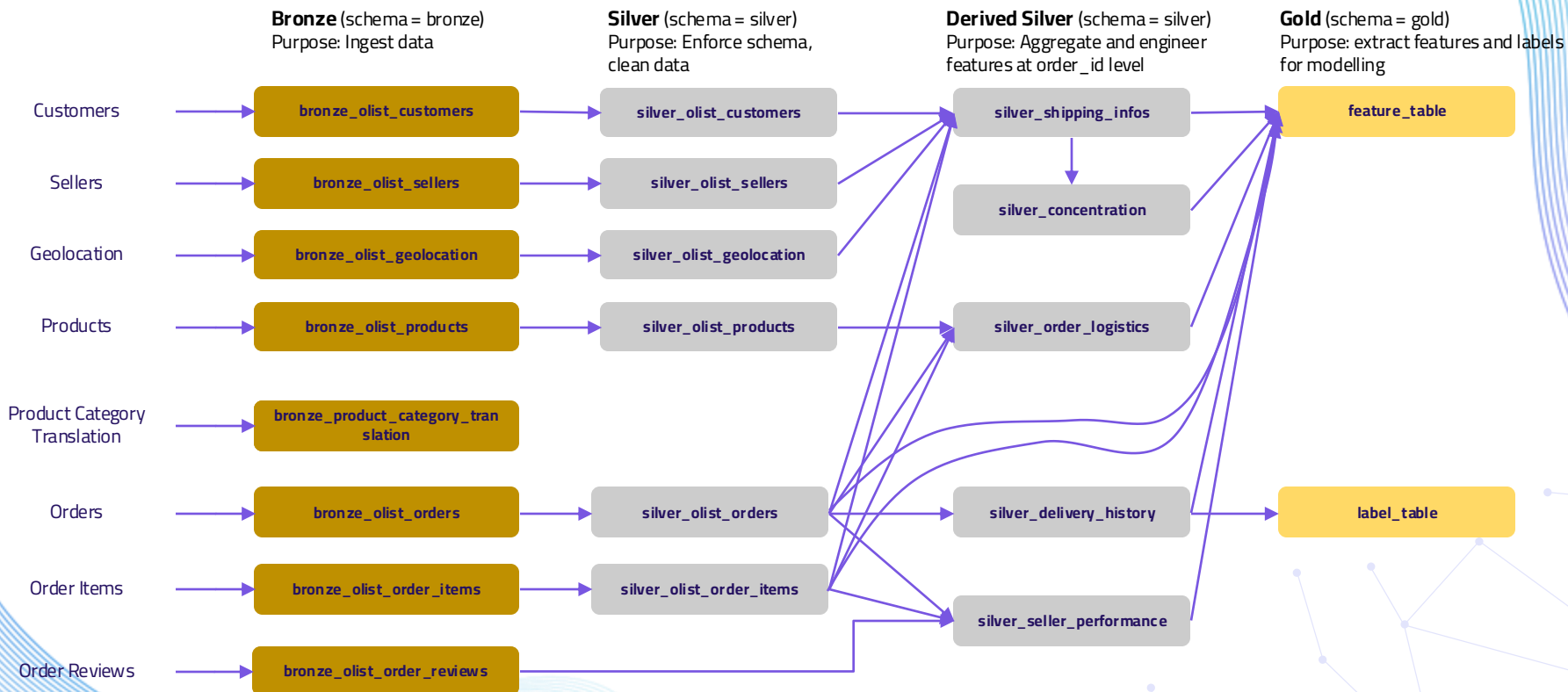


Solution Overview



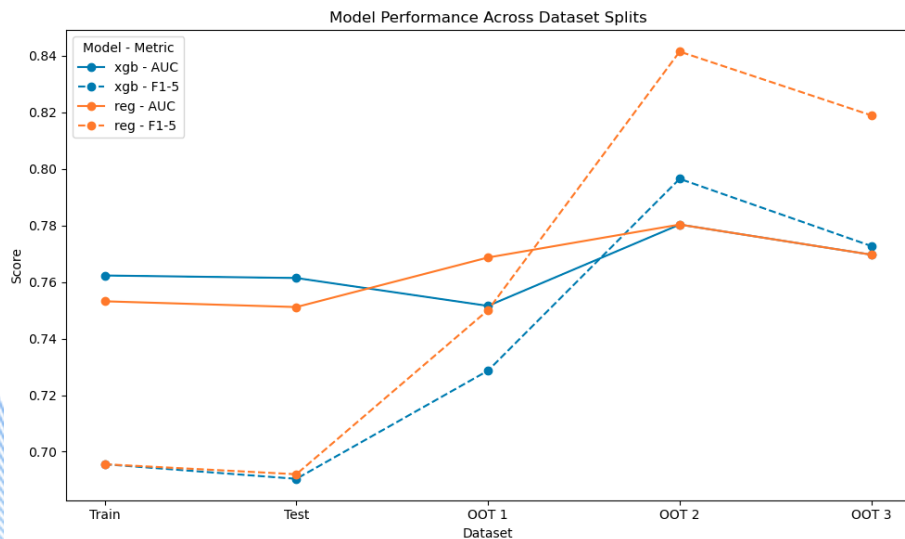
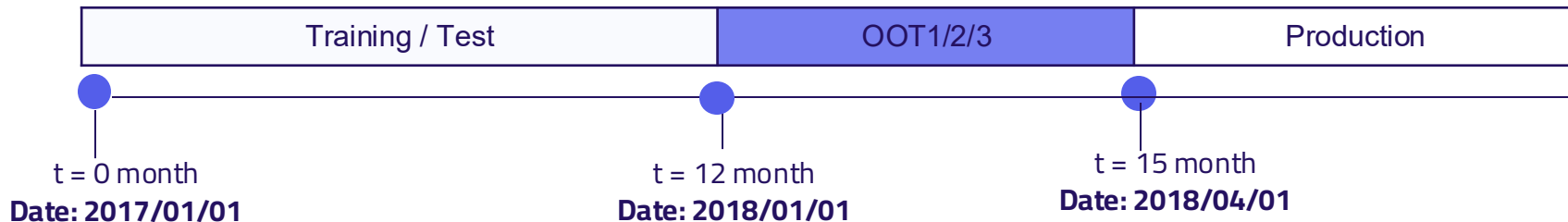
Data Processing Pipeline

- **Bronze Table:** Partitioned daily for orders to align with the prediction frequency
- **Derived Silver Tables:** Primarily partitioned at order level, leveraging **silverolist_orders** table as partitioning reference



Detailed schema can be found in our report

Model Training Strategy



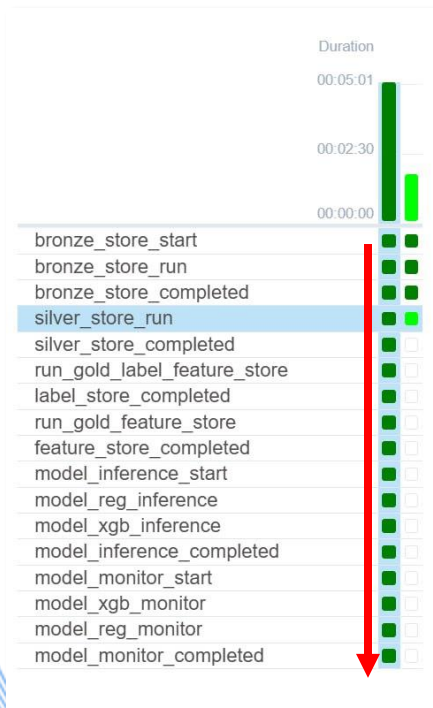
Data split strategy

- Train/Test - **12 months**: captures seasonality
- OOT - **3 months**: Validation across time

Results Summary: Regression model will be our primary model. Higher F1-5 accuracy across time.

Threshold: 0.25, the model will be aggressive in flagging likelihood of delays

Model inference: Batched daily at 9am



```
default_args = {
    'owner': 'airflow',
    'depends_on_past': True,
    'retries': 1,
    'retry_delay': timedelta(minutes=5),
}

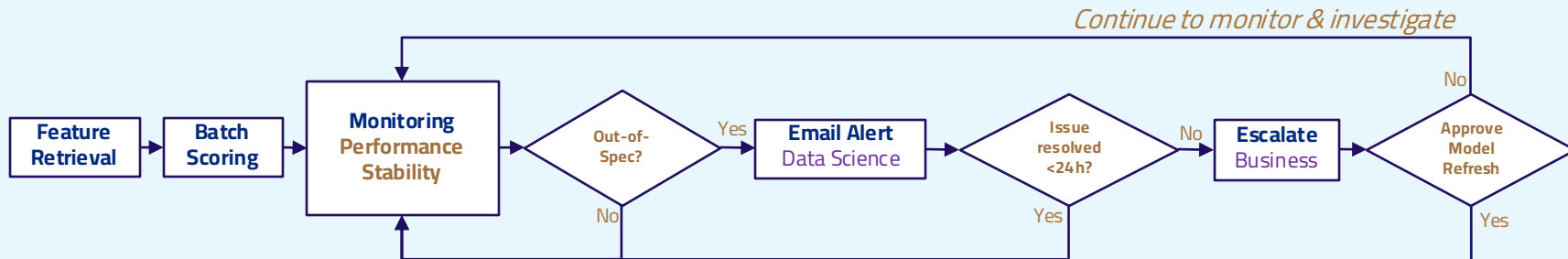
with DAG(
    'dag',
    default_args=default_args,
    description='Delivery lateness prediction
    pipeline',
    schedule_interval='0 9 * * *', # 09 AM
    daily
    start_date=datetime(2016, 9, 4), #min date
    end_date=datetime(2016, 10, 4), #'2018-09-
    03' max date
    catchup=True, # backfill if missing data on
    certain dates
    max_active_runs=1 # Execute all steps for
    day 1 then day 2
) as dag:
```

Key Design Considerations of Airflow DAG

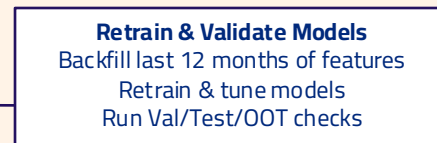
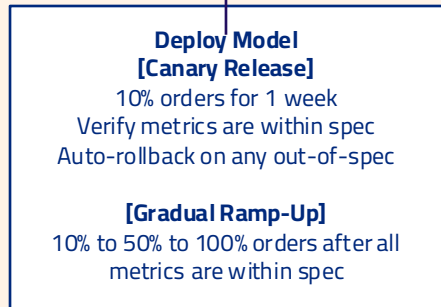
- **schedule_interval:** Daily at 9am to identify high risk deliveries before operations begin. Gives delivery team early warning to reroute and notify customers about expected delays
- **depends_on_past:** Provides safety, if job failed from previous day, will not continue, thus alerting engineers and biz team of the issue
- **max_active_runs:** Ensures data processing → model inference → monitoring tasks are sequential, mimicking real-life situation where data source is prepared first, followed by model inference and evaluation.

Model Monitoring & Re-Training SOP

Daily monitoring



Ad Hoc Retraining



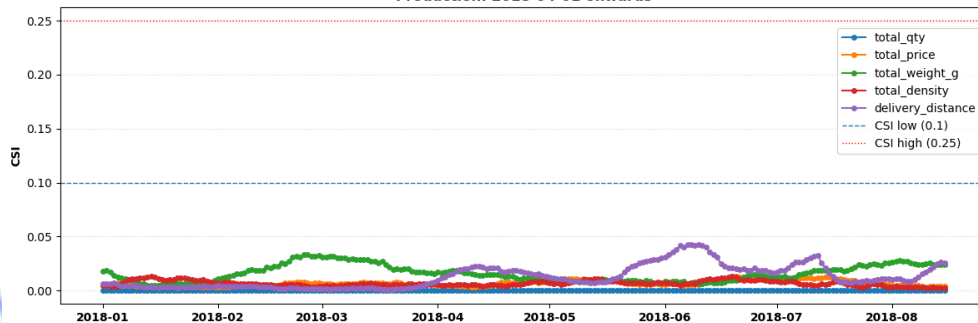
Scheduled Annual Retrain



Current Model Performance & Stability

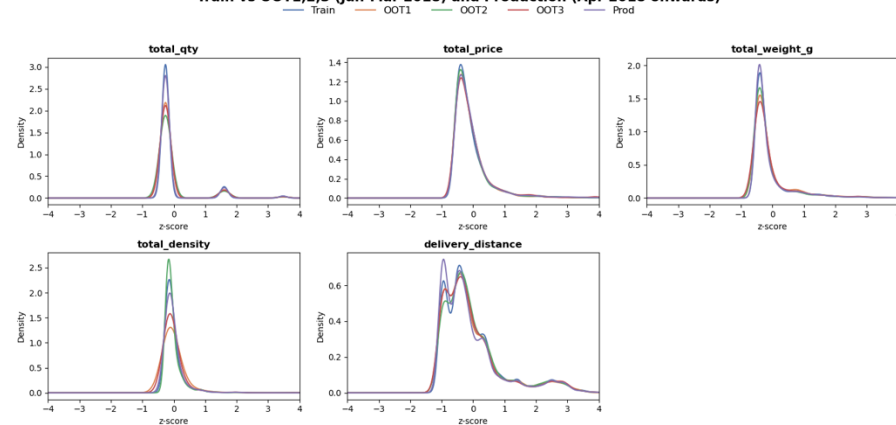
Characteristic Stability Index (CSI)

Baseline: 2017-01-01 to 2017-12-31
OOT Period: 2018-01-01 to 2018-03-31
Production: 2018-04-01 onwards



Feature Distribution

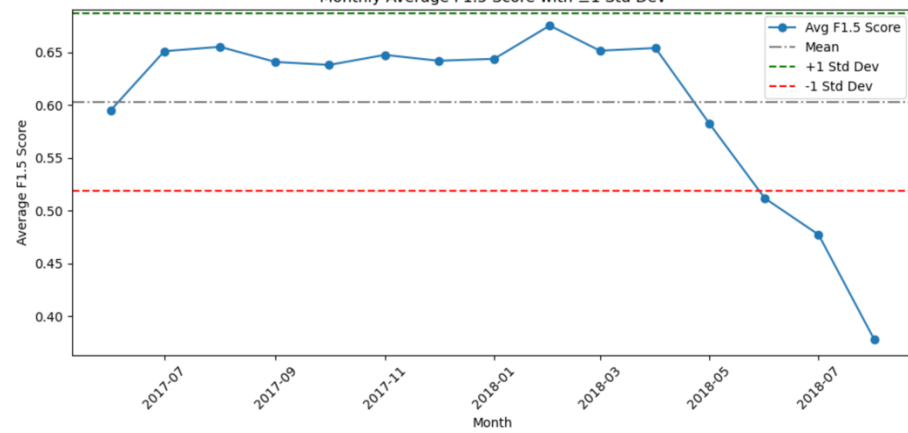
Train vs OOT1,2,3 (Jan-Mar 2018) and Production (Apr 2018 onwards)



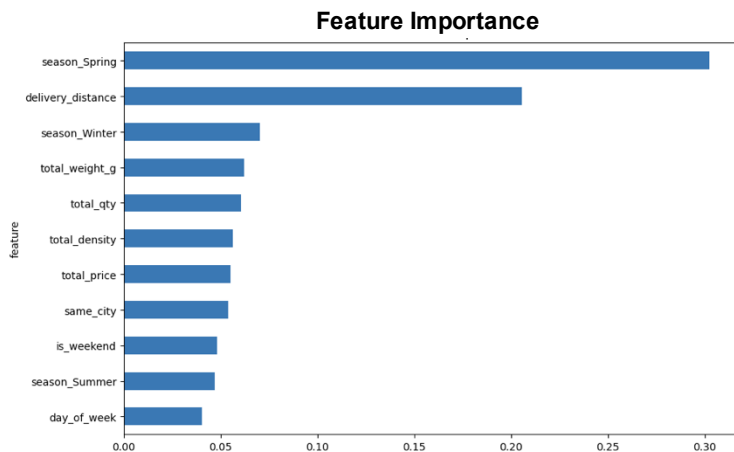
During OOTs and Production months,

- No signs of data drift for feature distributions
- All numeric features show stable CSI below <0.1
- F1.5 decreased after Production launch in April 2018. Performance was below our threshold of 0.525 in June 2018 and model retraining would be triggered.

Monthly Average F1.5 Score with ± 1 Std Dev



Key Insights & Next Steps for ML Pipeline



Distance is a risk factor

Longer delivery distance results in orders being **higher risks** (i.e. late)

Recommendation: Reduce seller-to-customer proximity by opening warehouses to serve more at-risk origin-destination routes



Season matters

Order volumes in Spring / Winter are **~50%** higher than Summer/Autumn

Recommendation: Proactively scale up logistics resources during peak seasons (Hire temp. staff or increase work hours)

Next Steps for Project:

- Enrich our ML pipeline with more data in these areas
- Improve operations with these insights while using delivery late prediction as a tool